1. Introduction

Networking is one of the fundamental areas of knowledge required by a Security Operations Center (SOC) analyst. Understanding how devices communicate, which protocols they use, and how data travels across a network helps analysts identify suspicious activities and investigate potential security incidents.

A SOC analyst must be able to distinguish legitimate network communication from unusual behavior. For example, an employee accessing a shared folder during working hours may be normal, whereas the same workstation attempting to access numerous internal systems late at night could require investigation.

This report documents my understanding of networking fundamentals, common protocols, DNS resolution, and how these concepts can be applied to network traffic analysis from a defensive security perspective.

2. Objectives of the Task

The objectives of this task were to:

- Understand the seven layers of the OSI model and how network communication occurs.
- Learn about encapsulation and decapsulation.
- Identify commonly used network ports and their associated protocols.
- Understand the differences between TCP and UDP.
- Explain the DNS resolution process.
- Recognize normal and potentially suspicious network traffic.
- Complete the assigned TryHackMe networking rooms.
- Apply networking knowledge to practical SOC investigation scenarios.

3. OSI Model

The Open Systems Interconnection (OSI) model is a conceptual framework that describes network communication through seven layers. Each layer has a specific responsibility and helps explain how data is transmitted between devices.

Layer	Name	Function
7	Application	Provides network services to applications; HTTP, DNS, SMB
6	Presentation	Data representation, encoding, compression and encryption
5	Session	Establishes and manages communication sessions
4	Transport	End-to-end transport using TCP or UDP
3	Network	Logical addressing and routing using IPv4 or IPv6
2	Data Link	Local network delivery using Ethernet frames and MAC addresses
1	Physical	Transmits bits as electrical, optical or radio signals

4. Common Network Ports and Protocols

Network ports help identify services running on devices and allow multiple applications to communicate over a network. Knowing common ports is important because security alerts frequently contain source and destination IP addresses, port numbers and protocol information.

Port	**Protocol**	**Purpose**
20/21	FTP	File transfer, including FTP control and traditional data connections
22	SSH	Secure remote administration
25	SMTP	Email transfer
53	DNS	Domain name resolution
80	HTTP	Unencrypted web communication
110	POP3	Email retrieval
143	IMAP	Email retrieval and mailbox access
443	HTTPS	Encrypted web communication
445	SMB	Windows file and printer sharing
3389	RDP	Remote desktop access

5. DNS Resolution Process

The Domain Name System (DNS) translates domain names into information such as IP addresses, allowing users and applications to locate network services without remembering numerical addresses.

When a user enters a domain name into a browser, the system attempts to resolve that name into an address that can be used to establish network communication.

Step-by-Step DNS Resolution

A simplified DNS resolution process is:

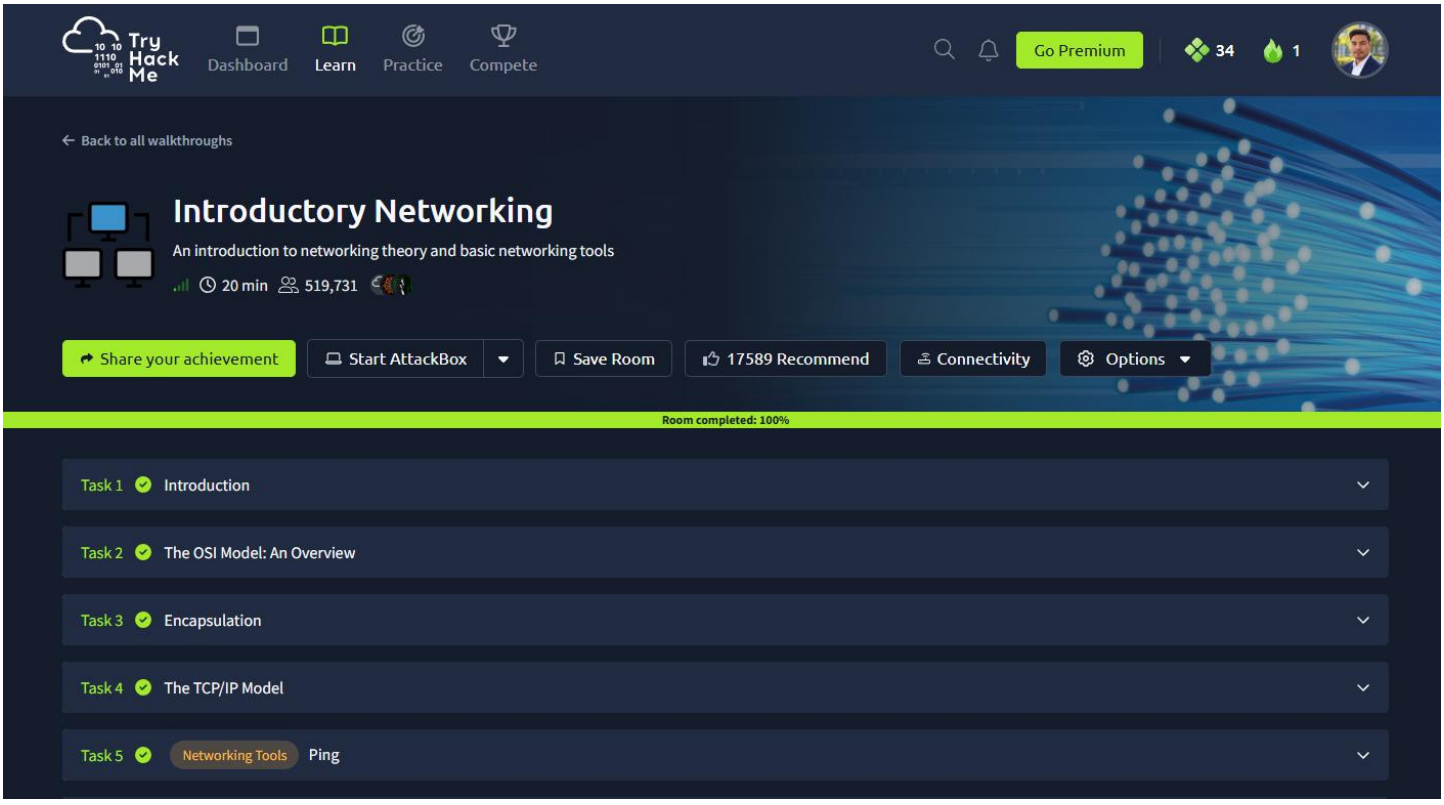
1. The device checks applicable local name-resolution sources, such as the hosts file and local DNS cache.
2. If the answer is unavailable locally, the device sends a DNS query to its configured resolver.
3. If the resolver does not already have the answer, it may perform recursive resolution or forward the request to another resolver.
4. During recursive resolution, a root DNS server can direct the resolver toward the relevant top-level domain (TLD) servers.
5. The TLD server can identify the authoritative name servers responsible for the requested domain.
6. The authoritative server provides the appropriate DNS response, such as an A or AAAA record.
7. The resolver returns the response to the client, and the answer may be cached according to applicable caching rules.

6. Practical Learning — TryHackMe

As part of this task, I completed three assigned TryHackMe rooms covering networking fundamentals.

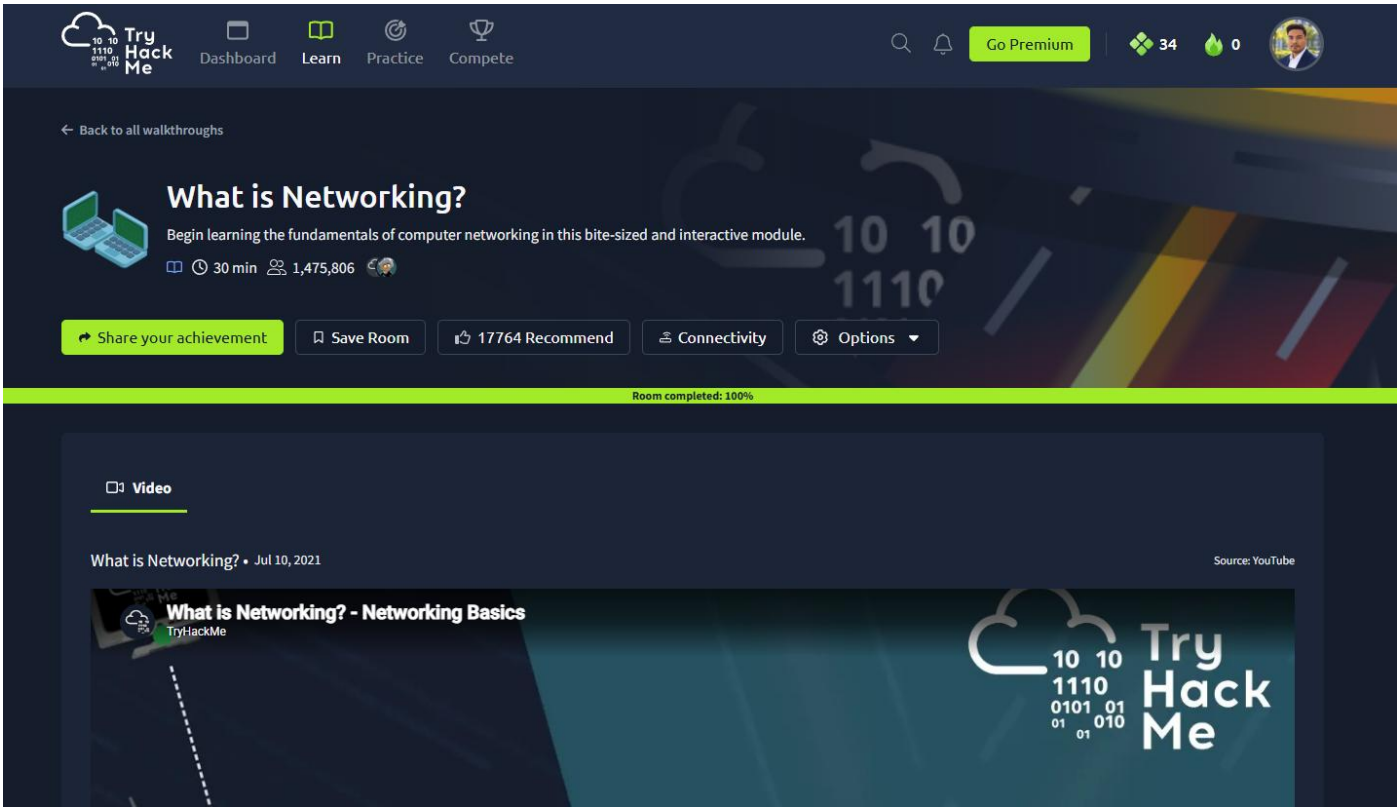
What is Networking?

This room helped reinforce foundational networking concepts and how devices communicate over networks.



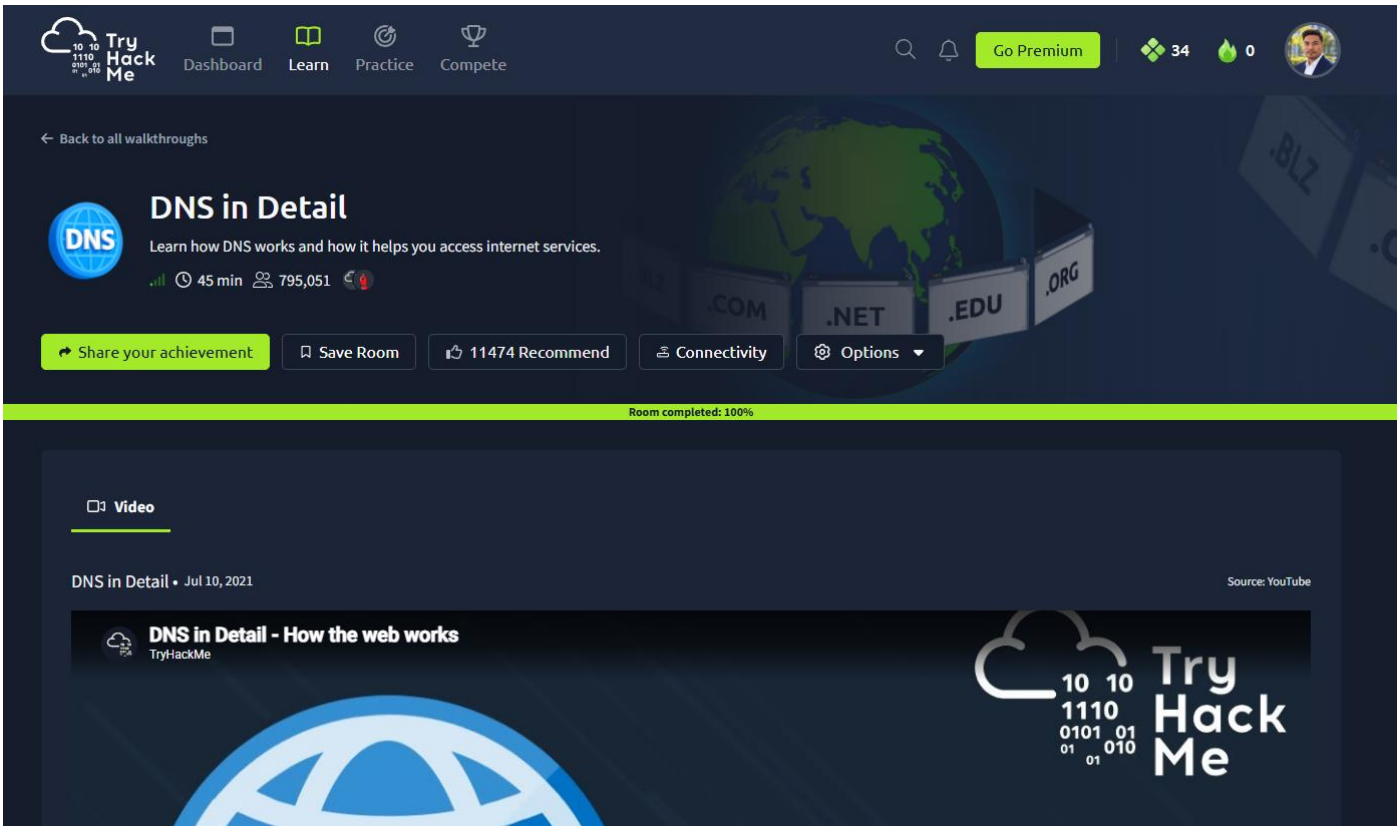
Introductory Networking

This room covered concepts including the OSI model, encapsulation, TCP/IP and network troubleshooting tools such as ping.



DNS in Detail

This room helped reinforce the DNS resolution process and the role of DNS infrastructure in translating domain names into network addresses.



7. Key Learning Outcomes

Through this task, I developed a stronger understanding of networking from a defensive security perspective.

My key learning outcomes include:

1. Understanding how the OSI model helps organize and interpret network communication.
2. Explaining encapsulation and the role of protocol headers in transmitting data.
3. Recognizing common network ports and their security relevance.
4. Differentiating between TCP and UDP communication.
5. Explaining the DNS resolution process.
6. Identifying patterns that may indicate SSH scanning, SMB lateral movement or DNS tunneling.
7. Understanding why suspicious traffic requires contextual investigation before reaching a conclusion.
8. Recognizing when correlated high-risk indicators justify prompt escalation.

8. Conclusion

Networking fundamentals are essential for understanding how security incidents can be identified through network activity.

This task helped me connect theoretical networking concepts with practical SOC investigation scenarios. I learned that identifying a protocol or port is only the beginning of an investigation. A SOC analyst must also consider communication patterns, user behavior, timing, authorization and supporting evidence.

The TryHackMe exercises and scenario-based investigations strengthened my understanding of how networking knowledge supports security monitoring, incident triage and escalation.

9. References and Supporting Evidence

TryHackMe learning resources

1. What is Networking? — https://tryhackme.com/room/whatisnetworking
2. Introductory Networking — https://tryhackme.com/room/introtonetworking
3. DNS in Detail — https://tryhackme.com/room/dnsindetail

Internship documentation

SOC Analyst Week 1 Task 2 — Notion Learning and Evidence Register:

https://app.notion.com/p/SOC-W01-T02-Networking-Fundamentals-from-a-Defender-s-View-416f4771e294817caad4fe2a5ee1b95e?source=copy_link